THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

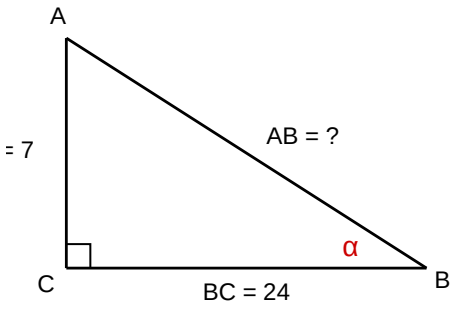
ANSWER KEY & MARKING SCHEME — MATHEMATICS · CLASS 10 · SET 03

Max. Marks: 80 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION – A (Multiple Choice Questions · 20 × 1 = 20) **20 questions · 20 marks**

Q1–Q18 are multiple-choice questions. Q19–Q20: a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

1. The value of tan α from the given figure is



(a) $\frac{24}{7}$ (b) $\frac{7}{25}$ (c) $\frac{7}{24}$ (d) $\frac{24}{25}$

> **Ans: (c)** tan α = AC/BC = 7/24.

2. ΔABC ~ ΔPQR. The perimeters of the two triangles are 30 cm and 45 cm. If AB = 12 cm, then PQ is

(a) 8 cm (b) 18 cm (c) 16 cm (d) 27 cm

> **Ans: (b)** Ratio of corresponding sides = ratio of perimeters = 2/3 ⇒ PQ = 12 × 3/2 = 18 cm.

3. The HCF and LCM of 8, 9 and 25 are respectively

(a) 1, 200 (b) 3, 1800 (c) 1, 900 (d) 1, 1800

> **Ans: (d)** The numbers are pairwise co-prime ⇒ HCF = 1, LCM = 8 × 9 × 25 = 1800.

4. Two cousins were both born in February 2024. The probability that they have the same birthday is

(a) $\frac{1}{29}$ (b) $\frac{1}{28}$ (c) $\frac{1}{366}$ (d) $\frac{2}{29}$

> **Ans: (a)** 2024 is a leap year, so February has 29 days ⇒ 1/29.

5. A point lies on the *y*-axis at a distance of 6 units from the origin. Which of the following can be its coordinates?

(a) (6, 0) (b) (−6, 6) (c) (0, −6) (d) (6, 6)

> **Ans: (c)** A point on the y-axis has x = 0; at distance 6 its y = ±6.

6. The value of $(2 + \sqrt{3})^2 + (2 - \sqrt{3})^2$ is

(a) 7 (b) $8\sqrt{3}$ (c) $4 + 7\sqrt{3}$ (d) 14

> **Ans: (d)** $(7 + 4\sqrt{3}) + (7 - 4\sqrt{3}) = 14$.

7. The number 7 × 11 × 13 + 13 is

(a) a composite number (b) a prime number
(c) neither prime nor composite (d) an irrational number

> **Ans: (a)** = 13(77 + 1) = 13 × 78, a product of two numbers greater than 1.

8. Rohit reads 12 pages on the first day and 4 pages more each day than on the previous day. The number of pages he reads on the 10th day is

(a) 44 (b) 48 (c) 52 (d) 40

Ans: (b) $a_{10} = 12 + 9 \times 4 = 48$.

9. The system $2x + 3y = 7$ and $(a + b)x + (2a - b)y = 21$ has infinitely many solutions. Then $a - b$ equals

(a) 6 (b) 5 (c) 1 (d) 4

Ans: (d) $2/(a+b) = 3/(2a-b) = 7/21 \Rightarrow a + b = 6$, $2a - b = 9 \Rightarrow a = 5, b = 1 \Rightarrow a - b = 4$.

10. Which of the following is NOT possible for an acute angle θ?

(a) $\sin\theta = \frac{4}{3}$ (b) $\tan\theta = 5$ (c) $\sec\theta = 1.2$ (d) $\cos\theta = 0.8$

Ans: (a) $\sin\theta$ of any angle lies between −1 and 1; 4/3 > 1.

11. The point P(x, y) divides the line segment joining A(−2, −2) and B(2, −4) in the ratio 3 : 7 internally. Then P is

(a) $(\frac{4}{5}, \frac{13}{5})$ (b) $(\frac{-4}{5}, \frac{13}{5})$ (c) $(\frac{-4}{5}, \frac{-13}{5})$ (d) $(\frac{-8}{5}, \frac{-6}{5})$

Ans: (c) $x = (3\times2 + 7\times(-2))/10 = -4/5$; $y = (3\times(-4) + 7\times(-2))/10 = -13/5$.

12. A 2-digit number is selected at random. The probability that it is a multiple of 3 and 5 but not of 2 is

(a) $\frac{1}{15}$ (b) $\frac{1}{30}$ (c) $\frac{1}{18}$ (d) $\frac{1}{10}$

Ans: (b) Two-digit numbers: 90. Odd multiples of 15: 15, 45, 75 ⇒ 3/90 = 1/30.

13. The solution of $2x + 3y = 12$ and $x - 3y = -3$ is

(a) x = 2, y = 3 (b) x = 6, y = 0 (c) x = 3, y = 2 (d) x = 1, y = 3

Ans: (c) Adding: $3x = 9 \Rightarrow x = 3$; then y = 2.

14. If x and y are distinct primes, the HCF of x^4y and x^2y^3 is

(a) x^2y (b) x^4y^3 (c) x^2y^3 (d) xy

Ans: (a) Smallest power of each prime: x^2y.

15. Two dice are thrown simultaneously. The probability that the product of the two numbers is 12 is

(a) $\frac{1}{12}$ (b) $\frac{1}{6}$ (c) $\frac{5}{36}$ (d) $\frac{1}{9}$

Ans: (d) (2,6), (6,2), (3,4), (4,3) ⇒ 4/36 = 1/9.

16. If $\operatorname{cosec}\theta = \frac{13}{12}$, then $\cot\theta$ equals

(a) $\frac{12}{5}$ (b) $\frac{5}{12}$ (c) $\frac{5}{13}$ (d) $\frac{13}{5}$

Ans: (b) $\sin\theta = 12/13 \Rightarrow \cos\theta = 5/13 \Rightarrow \cot\theta = 5/12$.

17. The pair of equations $3x + 2y = 7$ and $6x - y = 9$ represents two lines which are

(a) parallel (b) coincident
(c) intersecting at exactly two points (d) intersecting at exactly one point

Ans: (d) $3/6 = 1/2 \neq 2/(-1) \Rightarrow$ a unique solution.

18. The common difference of the AP $\sqrt{2}, \sqrt{8}, \sqrt{18}, \ldots$ is

(a) $2\sqrt{2}$ (b) $3\sqrt{2}$ (c) $\sqrt{2}$ (d) $4\sqrt{2}$

Ans: (c) $\sqrt{8} = 2\sqrt{2}$, $\sqrt{18} = 3\sqrt{2} \Rightarrow d = \sqrt{2}$.

19. **Assertion (A):** The points (2, 3), (4, 5) and (6, 7) are collinear.
Reason (R): If PQ + QR = PR for three points P, Q and R, then the points are collinear.

Ans: (a) PQ = QR = 2√2, PR = 4√2 = PQ + QR ⇒ collinear; R states the test used.

20. **Assertion (A):** The probability of an event always lies between 0 and 1, both inclusive.
Reason (R): For any event E, P(E) + P(not E) = 1.

Ans: (b) Both A and R are true, but R (P(E) + P(not E) = 1 with both non-negative) is not the standard explanation of the range 0 to 1 as stated in A; taken as in the sample key, R is a correct but separate statement ⇒ (b).

SECTION – B (Very Short Answer Questions · 5 × 2 = 10) — 5 questions · 10 marks

21. Find the HCF and LCM of 255 and 867 by the prime factorisation method.

255 = 3 × 5 × 17 and 867 = 3 × 17² (1); HCF = 3 × 17 = 51; LCM = 3 × 5 × 17² = 4335 (1).

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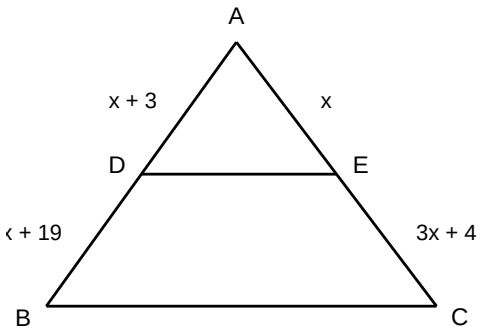
Find the smallest number which leaves remainder 2 when divided by 8, 12 and 15.

LCM(8, 12, 15) = 120 (1) ⇒ required number = 120 + 2 = 122 (1).

22. A flagpole 18 m high casts a shadow 9 m long. At the same time a building 30 m high casts a shadow of what length?

Similar triangles: 18/9 = 30/s (1) ⇒ s = 15 m (1).

23. In ΔABC, DE ∥ BC with AD = x + 3, DB = 3x + 19, AE = x and EC = 3x + 4. Find x.



BPT: (x + 3)(3x + 4) = x(3x + 19) (1) ⇒ 3x² + 13x + 12 = 3x² + 19x ⇒ x = 2 (1).

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The areas of two similar triangles are 36 cm² and 100 cm². If a side of the first triangle is 9 cm, find the corresponding side of the second.

Ratio of sides = √(36/100) = 6/10 (1) ⇒ side = 9 × 10/6 = 15 cm (1).

24. A kite is flying at an angle of elevation of 30° with the ground and the string attached to it is taut and 80 m long. Find the height of the kite above the ground.

sin 30° = h/80 (1) ⇒ h = 40 m (1).

25. The LCM of two numbers is 1200 and their HCF is 40. If one of the numbers is 240, find the other number.

Other = (HCF × LCM)/240 = (40 × 1200)/240 (1) = 200 (1).

SECTION – C (Short Answer Questions · 6 × 3 = 18) — 6 questions · 18 marks

26. Prove that 7√5 is irrational, given that √5 is irrational.

Assume 7√5 = p/q (q ≠ 0) rational (1) ⇒ √5 = p/(7q), a rational number (1), contradicting the given fact. Hence 7√5 is irrational (1).

27. Solve the pair of equations: $\frac{2}{x} + \frac{3}{y} = 13$; $\frac{5}{x} - \frac{4}{y} = -2$.

> Put p = 1/x, q = 1/y: 2p + 3q = 13; 5p − 4q = −2 (1). Eliminate: 8p + 12q = 52; 15p − 12q = −6 ⇒ 23p = 46 ⇒ p = 2, q = 3 (1). x = 1/2, y = 1/3 (1).

28. Which term of the AP 3, 8, 13, … is 78? Also find the sum of the terms up to that term.

> 78 = 3 + (n − 1)5 ⇒ n = 16 (1½). S_{16} = 16/2 (3 + 78) = 648 (1½).

OR

The sum of the first 7 terms of an AP is 49 and that of the first 17 terms is 289. Find the sum of the first *n* terms.

> 7/2 (2a + 6d) = 49 ⇒ a + 3d = 7; 17/2 (2a + 16d) = 289 ⇒ a + 8d = 17 (1½) ⇒ d = 2, a = 1. S_n = n/2 [2 + 2(n − 1)] = n^2 (1½).

29. Find the coordinates of the point P on the line segment joining A(−6, 10) and B(3, −8) such that P divides AB in the ratio 2 : 7.

> P = ((2×3 + 7×(−6))/9, (2×(−8) + 7×10)/9) (2) = (−4, 6) (1).

OR

Show that A(0, −1), B(−2, 3), C(6, 7) and D(8, 3) are the vertices of a rectangle.

> AB = CD = √20, BC = DA = √80 (1½); diagonals AC = BD = 10 (1). Opposite sides equal and diagonals equal ⇒ rectangle (½).

30. If $\tan\theta = \frac{3}{4}$, find the value of $\frac{\sin\theta - \cos\theta}{\sin\theta + \cos\theta}$.

> Take perpendicular 3, base 4, hypotenuse 5 (1). sin θ = 3/5, cos θ = 4/5 (1). Value = (3/5 − 4/5)/(3/5 + 4/5) = −1/7 (1).

31. A box contains 90 discs numbered 1 to 90. One disc is drawn at random. Find the probability that it bears (a) a two-digit number (b) a perfect square (c) a number divisible by 5.

> (a) 81/90 = 9/10 (1). (b) 1, 4, 9, 16, 25, 36, 49, 64, 81 ⇒ 9/90 = 1/10 (1). (c) 18/90 = 1/5 (1).

SECTION – D (Long Answer Questions · 4 × 5 = 20) — 4 questions · 20 marks

32. State and prove the converse of the Basic Proportionality Theorem.

> Statement: if a line divides two sides of a triangle in the same ratio, it is parallel to the third side (1). Given ΔABC, DE meets AB, AC with AD/DB = AE/EC; to prove DE ∥ BC (1). Suppose not; draw DE′ ∥ BC meeting AC at E′ (1). By BPT AD/DB = AE′/E′C. So AE/EC = AE′/E′C ⇒ (AE/EC) + 1 = (AE′/E′C) + 1 ⇒ AC/EC = AC/E′C ⇒ EC = E′C (1½), so E and E′ coincide ⇒ DE ∥ BC (½).

33. Meena withdrew ₹ 2000 from a bank in ₹ 50 and ₹ 100 notes and received 25 notes in all. How many notes of each kind did she receive?

> Let ₹50 notes = x, ₹100 notes = y: x + y = 25 (1); 50x + 100y = 2000 ⇒ x + 2y = 40 (1½). Subtract: y = 15 (1½), x = 10 (1). Check: 500 + 1500 = 2000 ✓.

OR

Solve graphically: 3*x* + *y* = 9 and *x* − *y* = −1. Find the points where the lines meet the *y*-axis.

> Tables: 3x + y = 9: (0, 9), (2, 3), (3, 0); x − y = −1: (0, 1), (2, 3), (−1, 0) (2). Graph (1½). Solution x = 2, y = 3 (1). y-axis points (0, 9) and (0, 1) (½).

34. From a point on a bridge across a river, the angles of depression of the two banks, on opposite sides of the river, are 30° and 45°. If the bridge is 3 m above the water, find the width of the river. (Take √3 = 1.732)

Let the point be A, 3 m above the water, and let the banks be B and C. Horizontal distance to one bank = 3/tan 30° = 3√3 m (2); to the other = 3/tan 45° = 3 m (2). Width = 3√3 + 3 = 3(1.732 + 1) = 8.196 m (1).

35. Prove that $\frac{\cos A - \sin A + 1}{\cos A + \sin A - 1} = \text{cosec } A + \cot A$, using the identity $\text{cosec}^2 A = 1 + \cot^2 A$.

Divide numerator and denominator by sin A: (cotA − 1 + cosecA)/(cotA + 1 − cosecA) (1½). Write 1 = $\text{cosec}^2 A - \cot^2 A$ in the numerator: (cotA + cosecA) − (cosecA − cotA)(cosecA + cotA) = (cotA + cosecA)(1 − cosecA + cotA) (2). Cancel ⇒ cosecA + cotA (1½).

OR

Prove that $\frac{1}{\text{cosec } \theta - \cot \theta} - \frac{1}{\sin \theta} = \frac{1}{\sin \theta} - \frac{1}{\text{cosec } \theta + \cot \theta}$.

1/(cosecθ − cotθ) = cosecθ + cotθ (multiply by conjugate; $\text{cosec}^2\theta - \cot^2\theta = 1$) (1½). So LHS = cosecθ + cotθ − cosecθ = cotθ (1). 1/(cosecθ + cotθ) = cosecθ − cotθ (1) ⇒ RHS = cosecθ − (cosecθ − cotθ) = cotθ (1½). LHS = RHS (1).

SECTION – E (Case Study Based Questions · 3 × 4 = 12) — 3 questions · 12 marks

36. A pottery workshop makes 12 pots on the first day. It increases production by 3 pots every day. The number of pots made on successive days forms an AP.

(i) State the first term and the common difference of the AP.

a = 12, d = 3 (1).

(ii) Find the number of pots made on the 10th day.

$a_{10} = 12 + 9 \times 3 = 39$ (1).

(iii) On which day will the workshop make 60 pots?

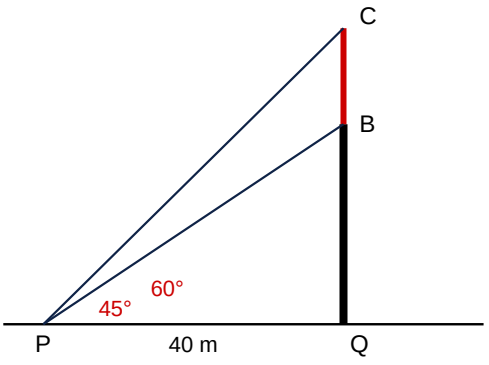
12 + (n − 1)3 = 60 ⇒ n − 1 = 16 ⇒ n = 17 (2).

OR

(iii) Find the total number of pots made in the first 12 days.

$S_{12} = 12/2\ [24 + 33] = 6 \times 57 = 342$ (2).

37. Students of Class X measure the height of their school building. They stand at a point P on level ground 40 m from the foot Q of the building. The angle of elevation of the top B of the building is 45°, and the angle of elevation of the top C of the flagpole fixed on the building is 60°. (Take √3 = 1.732)



(i) Find the height of the building BQ.

tan 45° = BQ/40 ⇒ BQ = 40 m (1).

(ii) Find the height of the point C above the ground.

tan 60° = CQ/40 ⇒ CQ = 40√3 = 69.28 m (1).

(iii) Find the length of the flagpole BC.

> BC = CQ − BQ = 40(√3 − 1) = 40 × 0.732 = 29.28 m (2).

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(iii) Find the distance of the point P from the top C of the flagpole.

> PC = √(40² + (40√3)²) = √(1600 + 4800) = 80 m (2).

38. In a lucky-draw box there are 30 cards bearing the numbers 1 to 30. A card is drawn at random.

(i) Find the probability that the number is a multiple of 6.

> 6, 12, 18, 24, 30 ⇒ 5/30 = 1/6 (1).

(ii) Find the probability that the number is a perfect cube.

> 1, 8, 27 ⇒ 3/30 = 1/10 (1).

(iii) Find the probability that the number is a prime or a multiple of 10.

> Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29 (10); multiples of 10: 10, 20, 30 (3); no overlap ⇒ 13/30 (2).

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(iii) The card numbered 15 is drawn and not replaced. Find the probability that the next card drawn is a multiple of 5.

> 29 cards remain; multiples of 5 left: 5, 10, 20, 25, 30 ⇒ 5/29 (2).